

GlocalDoc: A Global and Local Information Bottleneck for Few-Shot Legal Document Representation

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Abstract—Language models read a sentence well but lose the thread of a long document. For a court judgment this matters, because what a case is really about is spread across many paragraphs rather than stated in any one of them. We present GlocalDoc, a pre-training framework that teaches a paragraph-level encoder to read a judgment piece by piece and still hold on to the case as a whole. The idea is a teacher-student setup with a bottleneck: a momentum teacher encodes the clean judgment, a student encodes a corrupted copy, and a narrow learned channel forces the student to keep only what is needed to reproduce the teacher’s reading at three scales — each paragraph, a partial document, and the whole document. The global-alignment component follows the Glocal Information Bottleneck [1], proposed for time-series imputation, which we develop here for long, hierarchical documents. We evaluate GlocalDoc against a hierarchical masked language model baseline (H-MLM) and an unpretrained baseline on few-shot multi-label classification of European Court of Human Rights cases. Our working hypothesis was that a more global embedding should help most when supervision is scarce. The evidence supports this only in the extreme regime. With ten labelled examples per class GlocalDoc attains the best macro-F1 and, more tellingly, the lowest variance across seeds, while the masked language model often fails to learn at all. Once fifty or more examples per class are available the ordering reverses and H-MLM dominates. The bottleneck therefore behaves as a strong prior that is valuable under scarcity but discards lexical detail that becomes useful with more data. We also report a representational collapse specific to corpora with heavy boilerplate, and show that centring the teacher target removes it.

Index Terms—information bottleneck, self-distillation, document representation, few-shot learning, legal NLP, representational collapse.

I. Introduction

Today’s language models are very good at the small scale. Give one two sentences that mean the same thing and it returns nearly identical vectors, the property behind most retrieval and classification systems in use [2]. The trouble starts when the unit we care about is not a sentence but a whole, long document. A single forward pass sees only a few hundred tokens, so the usual fix — feed the model the opening and discard the rest — throws away most of the text. Gluing sentence vectors back together restores coverage but not structure: mean

or maximum pooling treats a judgment as a loose bag of paragraphs and forgets how those parts add up to a decision.

Legal text shows the problem at its sharpest, and it gives us the example we return to throughout the paper. Consider a European Court of Human Rights (ECtHR) judgment: an applicant is detained, the detention order cites no legal basis, and family visits are refused for months. Which Convention rights were violated is never written in any single paragraph — it emerges from how the detention facts and the visit facts sit together. Labelling such cases takes expert effort, so in practice one has a handful of examples per article, not thousands. What helps in that setting is a representation that already knows the overall shape of a document before it sees a single label, which is precisely what token-level pre-training does not give us.

We study whether an explicit compression objective can produce such a representation. The information bottleneck principle [3] prescribes keeping the part of an input that predicts a target while discarding the rest, and its variational form [4] makes the idea trainable inside a neural network. The specific move we build on — aligning the latent code of a masked input to that of its clean counterpart under a bottleneck — is the global-alignment principle of the Glocal Information Bottleneck [1], introduced for time-series imputation, where global guidance was shown to stop a model from overfitting local noise under heavy missingness. The same need arises for documents, where boilerplate and truncation play the role that missingness plays in time series; we adapt and extend the principle to a hierarchical, multi-granularity document setting in a framework we call GlocalDoc. A student encoder reads a corrupted view of a judgment and pools it into a single vector through a stochastic bottleneck; a momentum teacher reads the clean judgment; and the student is trained to predict the teacher at three granularities while paying a compression cost on the bottleneck. The collapse-avoiding asymmetry of non-contrastive self-distillation [5], [6] is what keeps the encoder from satisfying the alignment objectives with

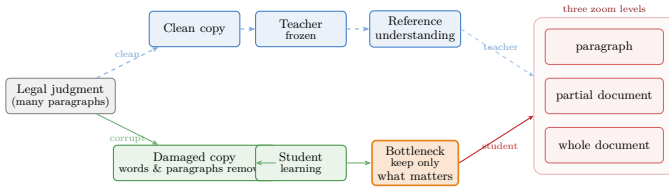


Fig. 1. The idea behind GlocalDoc. The student sees a damaged copy of a judgment, squeezes it through a bottleneck that keeps only what matters, and must still reproduce the teacher’s understanding at three zoom levels — a single paragraph, a partial document, and the whole judgment. Fig. 2 makes every component precise.

a constant output.

We compare GlocalDoc to H-MLM, a hierarchical masked language model in the spirit of SMITH [7] that keeps the same encoder and the same document pooler but replaces compression and alignment with token reconstruction plus a single pooling target. Both are evaluated by freezing the pre-trained encoder, then fine-tuning a linear classifier on $N \in \{10, 50, 100\}$ labelled cases per article. The questions we set out to answer are the following.

- RQ1. Does global and local bottleneck pre-training yield better few-shot document features than hierarchical masked language modelling?
- RQ2. How does any advantage depend on the amount of supervision available at fine-tuning time?
- RQ3. What training instabilities does the objective introduce on a boilerplate-heavy corpus, and which mechanisms resolve them?

The contributions of the paper follow these questions. We describe the GlocalDoc objective and the stabilisation mechanisms it needs in practice. We report a controlled comparison in which the only variable is the pre-training objective, and we find a clear interaction with N : GlocalDoc wins at $N = 10$ and H-MLM wins from $N = 50$ onward. We show that the most useful property of GlocalDoc at $N = 10$ is not a large mean improvement but a sharp reduction in seed variance, where the baseline collapses to zero on several seeds. Finally we document a failure mode that is easy to overlook on natural images but severe on legal text: when documents are nearly identical in the encoder’s initial embedding space, the teacher target degenerates to a constant, and centring it in the style of DINO [8] restores a usable training signal.

II. The Idea in Plain Terms

Before any equations, here is the whole method in one picture (Fig. 1). Picture two readers of the same court judgment. The teacher reads the clean, complete text and forms an impression of what the case is about. The student reads a damaged photocopy — some sentences smudged out, whole paragraphs missing — and has to arrive at the same impression. The catch is that the student must pass its understanding through a narrow bottleneck, so it

cannot simply copy surface wording. It has to keep only what genuinely matters about the case.

The student is then checked at three zoom levels and must reproduce the teacher’s reading at each. Up close: does every paragraph it produces match the teacher’s reading of that paragraph? At mid-range: from only the paragraphs that survived the damage, can it predict the teacher’s reading of the whole case? At the widest view: once everything has been squeezed through the bottleneck, does its single compressed code still recover the teacher’s whole-case reading? These three checks are the losses $\mathcal{L}_{\text{local}}$, $\mathcal{L}_{\text{inter}}$, and $\mathcal{L}_{\text{global}}$ of the next section. The same gist has to hold from close-up to wide-angle — which is what glocal, global and local together, is meant to capture.

Why a bottleneck rather than plain reconstruction? A model asked to rebuild every word tends to cling to local wording, and in boilerplate-heavy text that means clinging to noise. Making the student summarise through a narrow channel and match a global target instead works as a guardrail: it keeps the model honest about the case as a whole. This is the intuition the Glocal Information Bottleneck [1] introduced for time series, and it carries over to documents.

It helps to keep one case in mind. Return to the ECtHR judgment from the introduction: an applicant detained with no stated legal basis, family visits refused for months. The violated rights — liberty, and respect for family life — are never named in one place; they follow from how the detention and the visits combine. We use this case to make each mechanism concrete. Text masking smudges the sentence that names the missing legal basis. Paragraph dropout hides the visit-refusal paragraph and asks whether the student can still recover the family-life angle. Centring, introduced later, subtracts the identical procedural opening that every ECtHR judgment shares.

The rest of the paper turns each of these words — teacher, damaged copy, bottleneck, three-level match — into a precise, trainable object (Fig. 2). A reader who wants only the intuition can stop here and skip to the experiments.

III. The GlocalDoc Framework

This section turns the cartoon of Section II into precise objects. Figure 2 is the exact counterpart of Fig. 1 — the same two branches, the same bottleneck, the same three-level match, now with every tensor named. We fix notation, then describe the encoder, the two-level corruption, the bottleneck, the three alignment losses, and the mechanisms that keep training stable.

A. Notation and hierarchical encoding

We read a judgment the way one skims a case file: one paragraph at a time, then combine the per-paragraph impressions into a single sense of the whole. This subsection makes “impression” and “combine” precise. A document is a sequence of paragraphs $x = (p_1, \dots, p_N)$

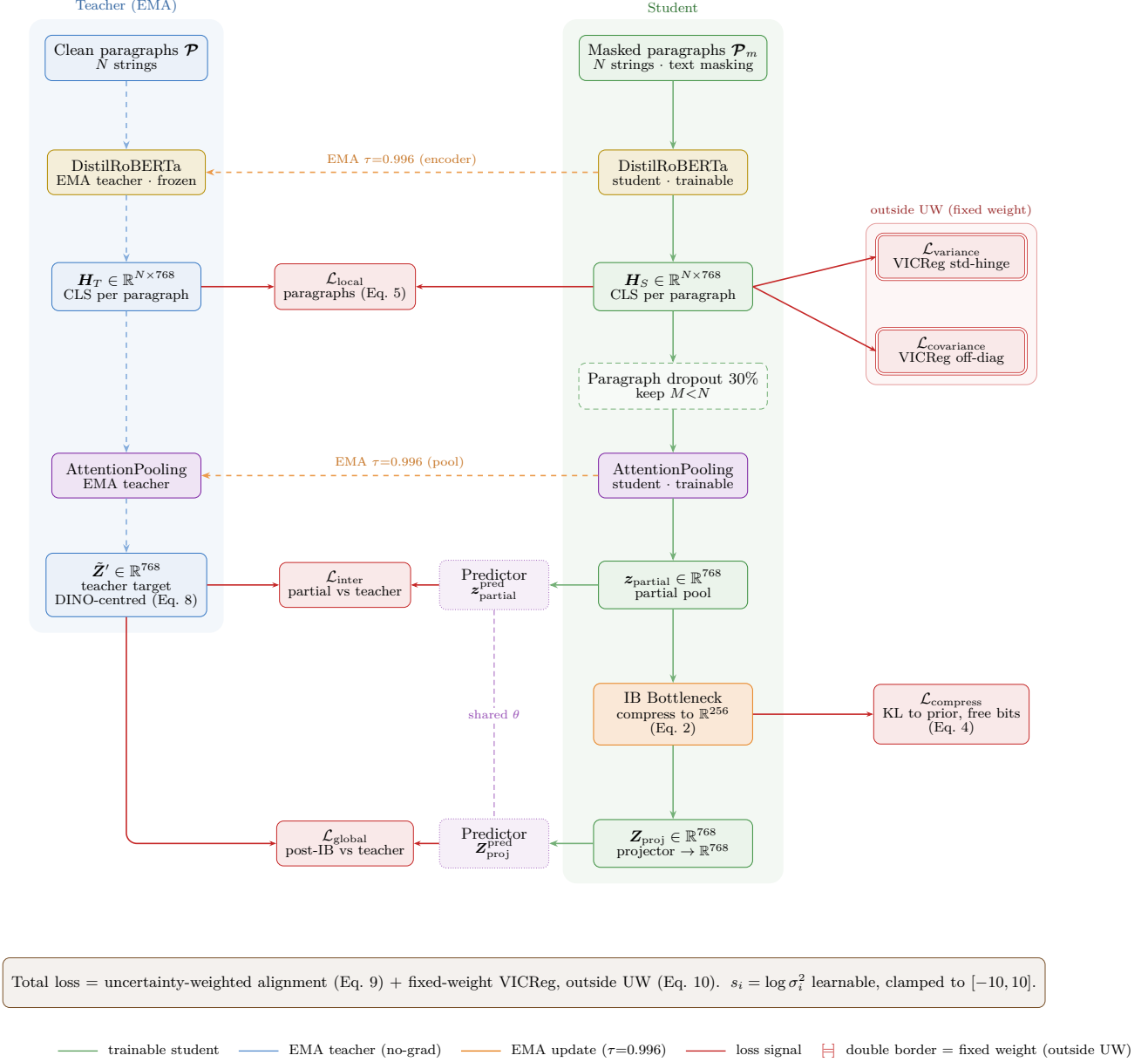


Fig. 2. The GlocalDoc architecture — the precise version of the intuition in Fig. 1. A frozen exponential-moving-average teacher (left, blue) encodes the clean paragraphs and produces a DINO-centred full-document target (Eq. 8). The trainable student (right, green) encodes a masked view, drops 30% of paragraphs at the pooling stage, compresses the partial pool through the information bottleneck (Eq. 2), and projects it back to $\mathbb{R}^{768}$. The central loss spine carries the three cosine alignment terms (Eq. 5): $\mathcal{L}_{\text{local}}$ matches paragraph vectors, while $\mathcal{L}_{\text{inter}}$ and $\mathcal{L}_{\text{global}}$ match the student’s partial- and full-document codes to the teacher target. A single student-only BYOL/SimSiam predictor (purple, $768 \rightarrow 512$, LayerNorm, GELU, $\rightarrow 768$, shared θ ; the teacher has none) breaks the constant-output fixed point. The compression term (Eq. 4) and the VICReg variance/covariance penalties (right lane, double border) carry fixed weights outside the uncertainty-weighted sum (Eqs. 9–10). Temporal VICReg on $Z_{\text{proj}}^{\text{pred}}$ and μ is applied at the script level and omitted here.

with $5 \leq N \leq 50$. Each paragraph is encoded independently by a shared transformer f_θ , taking the [CLS] representation as the paragraph vector. Paragraphs longer than 510 tokens are split into non-overlapping sub-chunks, each encoded separately, and their [CLS] vectors are averaged, so that a paragraph of any length maps to one vector in $\mathbb{R}^{768}$. Stacking gives the paragraph matrix $S = [s_1; \dots; s_N] \in \mathbb{R}^{N \times 768}$.

A document vector is produced by an attention pooler

$$a_i = \frac{\exp(q^\top (s_i + e_i))}{\sum_j \exp(q^\top (s_j + e_j))}, \quad \text{pool}(S) = \sum_i a_i (s_i + e_i), \quad (1)$$

where q is a learned query and e_i are learned positional embeddings. The positional embeddings are initialised to zero, so the pooler begins as exact mean pooling and departs from it only as training requires.

B. Two-level masking

This is the damaged photocopy of the Overview, made concrete: we corrupt the student’s input in two independent ways at once, so it has to recover the case from a partial, noisy view. For our running example, text masking might blur the sentence naming the missing legal basis, while paragraph dropout hides the visit-refusal paragraph altogether. GlocalDoc corrupts the input at two independent levels. At the text level, each paragraph is masked before being read by the student: sentences are dropped with probability 0.20, and within the surviving text spans of three to five words are replaced by a single mask token until about 15% of words are covered. At the structural level, a paragraph dropout removes 30% of paragraphs, but only at the pooling stage. The student encoder still sees all N masked paragraphs; the dropout selects a subset of the resulting vectors before pooling, which yields a partial view $z_{\text{partial}} = \text{pool}(S_{\text{kept}})$. Keeping all N paragraphs in the encoder is required so that the paragraph-level objective can compare student and teacher vectors position by position.

C. Information bottleneck

Here is the funnel. We push the student’s document vector through a deliberately narrow, slightly noisy channel, so only information that genuinely helps reproduce the teacher’s reading can get through. The compression cost is what keeps the channel narrow, and a warm-up stops it from biting before the student has learned anything worth keeping. Concretely, the partial document vector is passed through a stochastic bottleneck. Two linear heads produce a mean and a log standard deviation, and a latent code is sampled with the reparameterisation trick,

$$\mu = W_\mu z_{\text{partial}}, \quad \log \sigma = \text{clip}(W_\sigma z_{\text{partial}}, -10, 10), \quad (2)$$

$$z = \mu + \sigma \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (3)$$

with $z \in \mathbb{R}^{256}$. A two-layer projector g maps z back to $\mathbb{R}^{768}$, giving $Z_{\text{proj}} = g(z)$. The compression cost is the

Kullback-Leibler divergence to a standard normal prior, computed in closed form and floored per dimension by a free-bits budget λ ,

$$\mathcal{L}_{\text{compress}} = \beta_t \sum_{j=1}^{256} \max\left(\lambda, \frac{1}{2}(\mu_j^2 + \sigma_j^2 - 1 - 2 \log \sigma_j)\right). \quad (4)$$

The clip on $\log \sigma$ keeps the term finite under bfloat16. The weight β_t follows a linear warm-up from zero to one over the first quarter of training, which lets the alignment objectives organise the representation before compression starts to bite. We use $\lambda = 0.05$ nats per dimension; a larger floor instead pins the mean to a fixed magnitude regardless of the input, the opposite of what we want (Appendix A).

D. Multi-granularity alignment

These three losses are the three zoom levels of the Overview written as equations: at each scale, the student’s reading must match the teacher’s. The teacher is a momentum copy of the encoder and pooler, never updated by gradient. It reads the clean document and produces a full-document target $Z' = \text{pool}_T(S_T)$, which is centred as described below. Writing sg for stop-gradient, h for the predictor, and $d(u, v) = 1 - \cos(u, v)$ for the cosine distance, the three alignment losses are

$$\mathcal{L}_{\text{local}} = d(S, \text{sg}(S_T)), \quad (5)$$

$$\mathcal{L}_{\text{inter}} = d(h(z_{\text{partial}}), \text{sg}(Z')), \quad (6)$$

$$\mathcal{L}_{\text{global}} = d(h(Z_{\text{proj}}), \text{sg}(Z')). \quad (7)$$

The local term matches every student paragraph vector to the corresponding teacher paragraph vector. The intermediate term asks the partial pooled view to predict the full document before compression, and the global term asks the compressed and projected code to predict the same target after compression. Both document-level terms regress onto a single teacher vector, so the bottleneck must preserve whatever is needed to reconstruct the global representation from a partial and corrupted input. These two document-level terms are the document analogue of the global-alignment loss of the Glocal Information Bottleneck [1], with $\mathcal{L}_{\text{local}}$ playing the role of its local, point-wise counterpart at paragraph granularity.

E. Preventing collapse

A model can satisfy all three matching tasks by emitting the same vector for every document — a perfect score, and a useless representation. This subsection is the set of guardrails that close off that shortcut, and the decisive one turns out to be specific to legal text. The predictor h is applied to the student branch only. This asymmetry, together with the stop-gradient on the teacher, is the formal mechanism that removes the constant-output fixed point in non-contrastive self-distillation [5], [6]. The teacher tracks the student through an exponential moving average with rate $\tau = 0.996$, applied to both encoder

and pooler after every optimiser step; running statistics in normalisation layers are copied rather than averaged, since averaging them drifts away from the student’s activation distribution.

On its own the asymmetry was not enough on legal text. We add the variance and covariance regularisers of VICReg [9] on the student paragraph vectors: the first keeps each feature dimension from shrinking to a constant, the second decorrelates dimensions so they cannot collapse onto one another. Because per-document statistics cannot see variation across documents, we apply the same two penalties in a temporal form — against a rolling buffer of recent samples — to discourage a representation that is identical from one document to the next even when it varies within a document. The exact formulas are given in Appendix B.

The remaining and most important fix concerns the teacher target itself. ECtHR judgments share a large amount of boilerplate, and in the encoder’s initial space the documents are nearly indistinguishable: a probe over 200 documents found a median pairwise cosine of 0.9997, with every ordered pair above 0.99. A teacher target that is almost constant across documents lets the predictor satisfy $\mathcal{L}_{\text{inter}}$ and $\mathcal{L}_{\text{global}}$ by emitting that constant. We remove this fixed point with DINO-style centring [8]: a running mean c is maintained with momentum 0.9 and subtracted from the teacher output,

$$Z'_c = Z' - c, \quad c \leftarrow 0.9c + 0.1 \overline{Z'}_{\text{batch}}. \quad (8)$$

The centred vector is the actual regression target. Section VI shows that this single change is what keeps training stable.

F. Objective and optimisation

We are juggling four objectives at once, and hand-tuning four weights is fragile. Instead the model learns how much to trust each objective, while the anti-collapse guardrails are held at a fixed, non-negotiable weight. The four primary losses are combined by homoscedastic uncertainty weighting [10] with learned log-variances $s = (s_c, s_l, s_i, s_g)$,

$$\mathcal{L}_{\text{UW}} = \sum_k (e^{-s_k} \mathcal{L}_k + s_k), \quad (9)$$

and the anti-collapse regularisers are added with fixed weights outside the weighted sum,

$$\mathcal{L} = \mathcal{L}_{\text{UW}} + \omega_v V(S) + \omega_c C(S) + \mathcal{L}_{\text{temporal}}. \quad (10)$$

The weights s are given their own parameter group with a learning rate three orders of magnitude larger than the encoder and no weight decay. Without this, the gradient on s is too small to move the weights away from their initialisation, and the uncertainty weighting stays degenerate at uniform weights for the whole run.

G. The H-MLM baseline

To isolate what the bottleneck-and-alignment idea actually buys, we build a fair rival: it keeps the identical architecture but swaps the objective for the conventional one — predict masked tokens, plus a single pooling target — so any difference in results is attributable to the objective, not the network. H-MLM keeps the backbone, the paragraph encoding, and the attention pooler, and removes everything specific to GlocalDoc. It optimises a token-level masked language modelling loss $\mathcal{L}_{\text{mlm}}$ on each paragraph with a 15% mask rate, and a single pooling objective that asks the attention pool over a 70% subset of paragraphs to predict the stop-gradient mean of all paragraph vectors,

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{mlm}} + \frac{1}{2} d(\text{pool}(S_{\text{kept}}), \text{sg}(\overline{S})). \quad (11)$$

There is no bottleneck, no teacher, no predictor, and no uncertainty weighting. The comparison between GlocalDoc and H-MLM is therefore a comparison of pre-training objectives under an identical architecture and an identical fine-tuning procedure.

IV. Related Work

With the method in hand, we can place it among the ideas it builds on and the alternatives it is measured against.

A. Global-local information bottlenecks

Closest to our objective is the Glocal Information Bottleneck [1], proposed for time-series imputation. It augments an information bottleneck with a global-alignment loss — derived from a tractable approximation to the mutual information between a masked input’s representation and that of its clean original — alongside a local, point-wise reconstruction term. The method is model-agnostic and needs only a single MLP, and its motivation is that global guidance stops a model from overfitting local noise when much of the input is missing. GlocalDoc inherits this global-versus-local split directly: our $\mathcal{L}_{\text{inter}}$ and $\mathcal{L}_{\text{global}}$ are the global-alignment idea applied to documents, and $\mathcal{L}_{\text{local}}$ is its local counterpart at paragraph granularity. What we add is everything that carries the idea from time series to long, structured legal text: a hierarchical paragraph encoder, two-level masking, an intermediate partial-document level, an EMA teacher with a student-only predictor, DINO-style centring against boilerplate, and a few-shot evaluation that exposes a crossover with the amount of supervision. Table I summarises the split between what is inherited and what is new.

B. Information bottleneck and variational compression

The information bottleneck frames representation learning as a trade-off between keeping information about a target and compressing the input [3], [11]. The deep variational bottleneck makes this practical by parameterising a stochastic encoder and bounding the mutual

TABLE I
What GlocalDoc inherits from the Glocal Information Bottleneck [1] and what it adds for hierarchical legal documents.

Component	Source
Global-alignment loss (masked code vs. clean code under an IB)	Glocal-IB
Local, point-wise alignment term	Glocal-IB
Variational bottleneck, free bits, β warm-up	VIB / Glocal-IB
Hierarchical paragraph encoder + attention pooling	ours
Two-level masking (text spans + paragraph dropout)	ours
Intermediate, partial-document alignment level	ours
EMA momentum teacher + student-only predictor	ours, via BYOL/DINO
DINO-style centring for boilerplate collapse	ours
Few-shot legal evaluation + N -dependent crossover	ours

information terms [4], which connects it directly to the variational autoencoder [12] and to β -VAE [13], where a weight on the Kullback-Leibler term controls the strength of compression. A recurring difficulty is posterior collapse, in which the encoder ignores the input and the bottleneck reverts to its prior. The free-bits heuristic [14] addresses this by exempting a small budget of information per latent dimension from the compression penalty. GlocalDoc uses a hard per-dimension form of free bits together with a warm-up on β , so that compression is introduced only after the alignment objectives have shaped the representation.

C. Non-contrastive self-distillation

Self-supervised methods that align two views of an input without negative pairs must avoid the trivial solution of a constant embedding. BYOL [5] and SimSiam [6] achieve this with an asymmetric design: a predictor network on one branch, a stop-gradient on the other, and in BYOL a slow moving-average (momentum) teacher in the lineage of MoCo [15]. DINO [8] adds centring and sharpening of the teacher output to keep it from concentrating on a single direction. VICReg [9] and Barlow Twins [16] take a complementary route and prevent collapse explicitly by regularising the variance and the cross-correlation of the embeddings. GlocalDoc combines these ideas rather than choosing among them, because on our corpus each one alone proved insufficient, a point we return to in Section VI and Appendix B.

D. Masked language models and long documents

BERT [17] and RoBERTa [2] established masked language modelling as the dominant pre-training objective for encoders, and DistilRoBERTa [18] provides a compact six-layer version that we adopt as the shared backbone. None of these handle long documents natively. Longformer [19] extends the context window with sparse attention, whereas hierarchical models such as SMITH [7] encode sentences or paragraphs independently and compose them

with a second-level module. Our H-MLM baseline is hierarchical in this sense: it encodes each paragraph, trains an attention pooler to reconstruct the document from a partial view, and keeps token-level masked prediction as an auxiliary loss.

E. Legal NLP and few-shot classification

LexGLUE [20] collects several legal benchmarks, including the ECtHR allegation task derived from earlier work on judgment prediction [21]. The task is multi-label and strongly imbalanced, which makes macro-averaged F1 the appropriate primary metric. Few-shot text classification has been studied extensively for in-context learning with large decoders [22], but the encoder-plus-linear-head setting we use isolates the quality of the pre-trained representation from any prompting effect. Document and sentence embedding methods such as Sentence-BERT [23] and SimCSE [24] optimise pooled vectors directly, yet they target short inputs and do not model paragraph structure, which is the gap GlocalDoc addresses.

F. Balancing multiple objectives

GlocalDoc optimises four losses at once. Rather than searching for fixed weights we use homoscedastic uncertainty weighting [10], which treats each loss as a Gaussian likelihood with a learned observation noise and produces self-tuning weights. We keep the anti-collapse regularisers outside this mechanism, since a signal whose purpose is to prevent degeneracy must not be down-weighted by the same optimiser that benefits from degeneracy.

V. Experimental Setup

A. Data

We use the ECtHR Task A subset of LexGLUE [20], in which the input is the list of fact paragraphs of a case and the target is the set of allegedly violated Convention articles over ten labels. We keep documents with between five and fifty paragraphs, which removes fewer than one percent of cases and bounds the cost of encoding. The label distribution is severely skewed. The most frequent article appears in roughly 4,700 training cases and the rarest in about 41, a ratio close to one hundred. This motivates macro-averaged F1 as the primary metric, with micro-averaged F1 reported alongside it.

B. Pre-training

Both objectives pre-train DistilRoBERTa [18] on the same filtered training split. GlocalDoc runs for five epochs and H-MLM for three, both with batch size one and gradient accumulation, in bfloat16 with a linear schedule and ten percent warm-up. The full configuration is listed in Appendix C, Table III. We deliberately report a single pre-training run per objective, which is a limitation we discuss in Section VII.

TABLE II

Few-shot test results on ECtHR Task A. Mean over five seeds, standard deviation in parentheses. Best per row block in bold; the best method at each N is the bold value within that N .

Method	N	Macro-F1	Micro-F1
No pre-training	10	0.001 (0.001)	0.003 (0.004)
	50	0.025 (0.008)	0.077 (0.030)
	100	0.042 (0.013)	0.106 (0.019)
H-MLM	10	0.019 (0.023)	0.069 (0.085)
	50	0.273 (0.061)	0.354 (0.060)
	100	0.372 (0.032)	0.454 (0.030)
GlocalDoc	10	0.044 (0.009)	0.139 (0.045)
	50	0.108 (0.024)	0.224 (0.027)
	100	0.183 (0.006)	0.306 (0.014)

C. Fine-tuning protocol

At fine-tuning time the pre-trained encoder and pooler are loaded into a document classifier with a linear head over the pooled vector. For GlocalDoc we load the student pooler, since the teacher pooler is a moving average that barely departs from its initialisation over a short run. For each condition we evaluate $N \in \{10, 50, 100\}$ labelled cases per article, sampled with a multi-label aware routine that keeps the per-article quota and deduplicates cases covering several articles. Each setting is repeated over five seeds. The seed is fixed before both the few-shot sampling and the construction of the classifier head, so that a seed varies both the sample and the head initialisation rather than the sample alone. We train for ten epochs with a binary cross-entropy loss over the ten labels, a cosine schedule with ten percent warm-up, and a decision threshold of 0.5 at evaluation. The checkpoint with the best validation macro-F1 is evaluated on the held-out test split.

D. Baselines

Besides H-MLM, we include a no-pre-training baseline that loads an off-the-shelf DistilRoBERTa with a freshly initialised pooler. It measures how much of the few-shot performance is attributable to pre-training at all, as opposed to the architecture and the fine-tuning procedure.

VI. Results and Analysis

Table II reports the test results and Figure 3 plots the macro-F1 against N . We read them against the three research questions.

A. RQ1 and RQ2: the advantage is regime-dependent

There is no single winner. At $N = 10$ GlocalDoc reaches a macro-F1 of 0.044 against 0.019 for H-MLM and 0.001 for the unpretrained baseline, and the same ordering holds for micro-F1. From $N = 50$ the picture inverts: H-MLM reaches 0.273 against 0.108 for GlocalDoc at $N = 50$, and 0.372 against 0.183 at $N = 100$. The hypothesis that a global bottleneck representation helps under scarcity is therefore confirmed only in the smallest regime, and is

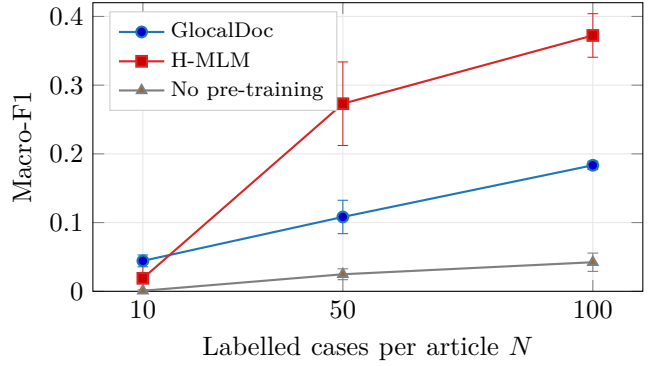


Fig. 3. Macro-F1 against the number of labelled cases per article, with one standard deviation over five seeds. GlocalDoc leads at $N = 10$ and has the smallest spread there; H-MLM overtakes it from $N = 50$ and widens the gap at $N = 100$.

contradicted once a moderate amount of supervision is available.

The most informative quantity at $N = 10$ is the variance rather than the mean. The per-seed scores in Appendix D, Table IV, show that H-MLM returns 0.000 and 0.001 on two of five seeds at $N = 10$, that is, it fails to learn anything on those runs, while GlocalDoc stays in a narrow band between 0.035 and 0.057. The bottleneck behaves as a strong prior. When ten cases per article are not enough to identify a good linear separator from a high-variance MLM representation, the compressed representation still supports a usable one, and it does so consistently. As N grows the prior turns into a handicap, because the same compression discards lexical detail that the MLM objective preserves and that a linear head can exploit once it has enough labelled examples to do so. The unpretrained baseline stays near zero throughout, which confirms that the effects we measure come from pre-training and not from the architecture or the optimiser.

B. RQ3: collapse and the role of centring

GlocalDoc does not train out of the box on this corpus. Before the centring fix, the inter-document cosine of the student document code climbed to about 0.99 within the first few hundred steps and the run produced the degenerate representation already visible in the early baseline numbers. Figure 4 shows the first epoch after the fix. The collapse metric, defined as the mean pairwise cosine of the post-predictor document codes over a sliding window, falls from 0.42 to below 0.05 and stays well under the alarm and stop thresholds for the rest of the epoch (panel a). The centred teacher target has a pairwise cosine near zero throughout, which is direct evidence that centring absorbs the constant component of the boilerplate-dominated representation. The bottleneck stays informative: the fraction of latent dimensions above the free-bits floor rises toward 0.58 as β warms up, and the mean magnitude of μ grows in step (panel b). The two document-level alignment losses decrease over the

epoch despite the rising compression weight (panel c). The uncertainty weighting is active rather than degenerate (panel d): the log-variance of the compression term grows, which down-weights a loss whose floor is already handled by free bits, while the local term is up-weighted as the hardest of the four to satisfy. Without centring none of this happens, and the same configuration collapses within the first ten percent of training.

C. Document homogeneity

The centring requirement is a property of the corpus, not of the method in general. The probe reported in Section III, a median pairwise cosine of 0.9997 over 200 documents, quantifies how little the raw encoder distinguishes one judgment from another. On a corpus whose documents already occupy distinct regions of the embedding space, the teacher target carries enough variation on its own and the predictor asymmetry alone would likely suffice. Legal text is at the opposite extreme, and we expect any non-contrastive document objective to need an explicit centring or sharpening step on such data.

VII. Discussion

The result that organises the paper is the crossover in Figure 3. A compression objective and a reconstruction objective produce representations with different inductive biases, and the better choice depends on how much supervision the downstream task will receive. Under ten examples per article the compressed representation is more linearly separable and far more stable, which is exactly what a low-data practitioner needs. Past fifty examples the reconstruction representation provides richer features that a linear head can exploit, and the compression that helped earlier now removes useful signal. Reading the bottleneck as a learned prior makes both halves of this behaviour expected rather than contradictory.

The stability finding deserves its own emphasis. Mean macro-F1 at $N = 10$ is low for every method, as it must be with ten examples and ten imbalanced labels, but the spread differs sharply. A method that returns zero on forty percent of seeds is not usable in a setting where one cannot afford many annotation rounds, and on this axis GlocalDoc is clearly preferable even where the mean gap to H-MLM is small.

Several limitations bound these conclusions. We report one pre-training run per objective, so the variance we measure is over fine-tuning seeds and not over pre-training randomness; a pre-training seed could move the crossover point. The backbone is a distilled six-layer encoder and the pre-training is short, which keeps the experiments affordable but leaves open whether a larger model or a longer schedule would let the bottleneck retain more of the detail it currently discards. The stabilisation mechanisms were introduced as a sequence of diagnostic interventions against observed collapse rather than as a controlled ablation, so we can state that the combination trains and

that centring is the decisive component on this corpus, but we cannot yet attribute precise contributions to the variance, covariance, and temporal terms. All results are on a single dataset in a single legal jurisdiction.

VIII. Conclusion

We introduced GlocalDoc, a pre-training framework that compresses a corrupted, partially observed document through an information bottleneck while aligning it to a momentum teacher at the paragraph, partial-document, and full-document level. On few-shot classification of ECtHR cases the framework is preferable to a hierarchical masked language model only in the smallest-data regime, where it is both more accurate and far more stable across seeds, and it is outperformed once fifty or more labelled cases per article are available. We traced a representational collapse to the near-constant teacher target produced by boilerplate-heavy legal text, and showed that centring the target removes it. Future work should run controlled ablations of the stabilisation terms, test a decoupled or annealed compression schedule that preserves more lexical detail at larger N , and study whether the crossover point can be shifted with a larger backbone, a finer-grained sweep of the number of labelled examples around the observed crossover, or a curriculum over it.

Appendix A Loss details

The compression term is the closed-form Kullback-Leibler divergence between the diagonal Gaussian posterior $\mathcal{N}(\mu, \text{diag}(\sigma^2))$ and the standard normal prior, computed from $\log \sigma$ for numerical stability under bfloat16. The per-dimension free-bits floor replaces each term by its maximum with λ , so a dimension contributes no gradient while it uses fewer than λ nats. With $\lambda = 0.05$ and a 256-dimensional latent the floor on the total divergence is 12.8 nats, low enough to permit substantial compression while ruling out a full posterior collapse to the prior. A larger floor over-constrains the model: at $\lambda = 0.5$ the resulting 128-nat floor dominates the objective and pins the mean to a fixed magnitude regardless of the input, which is why we settled on $\lambda = 0.05$. The cosine objectives use $d(u, v) = 1 - \cos(u, v)$; $\mathcal{L}_{\text{local}}$ flattens the per-document paragraph matrices before computing the mean cosine, so each student paragraph is matched to the teacher paragraph at the same position. The uncertainty weighting follows the homoscedastic form of [10], where each learned s_k is the log-variance of a Gaussian likelihood and the term s_k acts as a regulariser that prevents the weights from growing without bound.

Appendix B Diagnosing collapse

The stabilisation mechanisms were added one at a time in response to distinct failure modes. A shared stop-gradient encoder collapsed because it could satisfy every

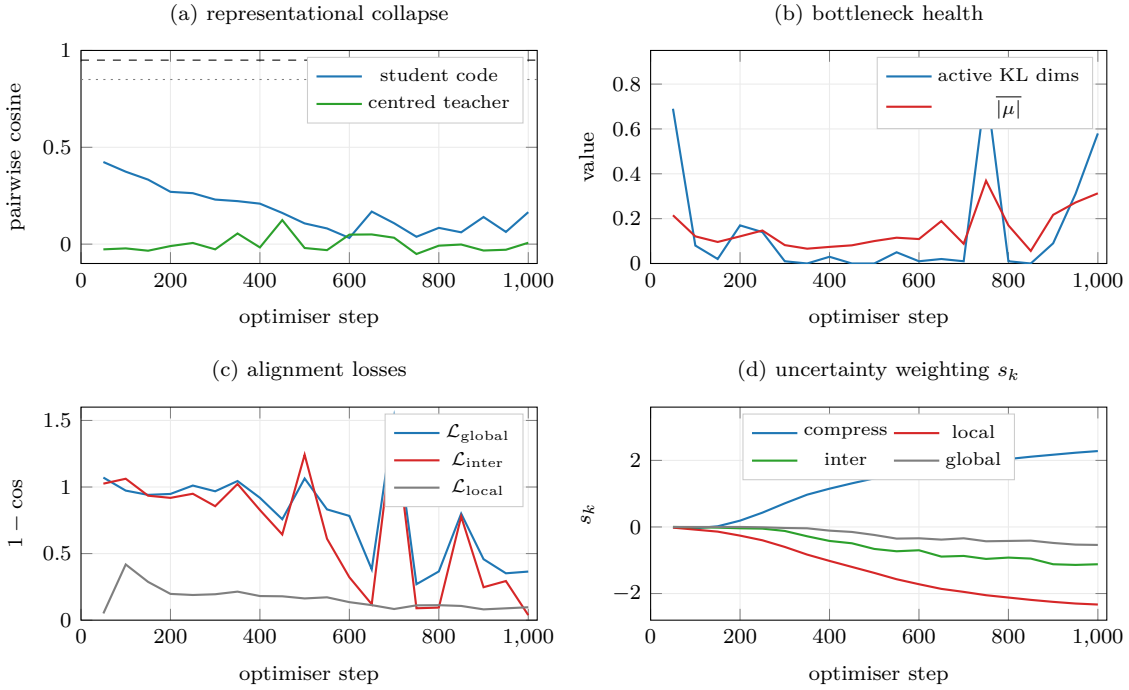


Fig. 4. First epoch of GlocalDoc pre-training after the centring fix. (a) The collapse metric stays far below the alarm (0.85, dotted) and stop (0.95, dashed) thresholds, and the centred teacher target has near-zero pairwise cosine. (b) The bottleneck remains informative as β warms up, with the fraction of active latent dimensions and the mean magnitude of μ both rising. (c) The document-level alignment losses decrease despite the growing compression weight. (d) The uncertainty weighting log-variances separate, which shows the weighting is active rather than stuck at its uniform initialisation.

alignment loss by becoming input-invariant; replacing it with a separate momentum teacher encoder and a student-side predictor removed that fixed point and let the encoder move. The bottleneck then collapsed downstream, with diverse paragraph vectors mapped to a constant document code; temporal variance and covariance terms on the document code and on the bottleneck mean addressed this. Aggressive free bits at $\lambda = 0.5$ over-constrained the mean and were reduced to $\lambda = 0.05$. The decisive failure was the near-constant teacher target induced by boilerplate, measured as a median inter-document cosine of 0.9997, which DINO centring removed. The first stable epoch is the one shown in Figure 4.

The variance and covariance regularisers applied to the student paragraph matrix S are

$$V(S) = \frac{1}{d} \sum_j \max(0, \gamma - \sqrt{\text{Var}(S_{:,j}) + \epsilon}), \quad C(S) = \frac{1}{d} \sum_{i \neq j} [\text{Cov}(S_{:,i}, S_{:,j})] \quad (12)$$

with variance hinge $\gamma = 0.5$. The temporal variant applies the same two expressions to the post-predictor document code $\mathbf{Z}_{\text{proj}}^{\text{pred}}$ and the bottleneck mean μ , computed against a detached rolling buffer of the sixteen most recent samples, so that $\mathcal{L}_{\text{temporal}}$ in Eq. 10 penalises representations that stay constant across consecutive documents.

Appendix C Hyperparameters

Table III lists the configuration of both pre-training objectives and the shared fine-tuning procedure.

Appendix D Per-seed results

Table IV gives the macro-F1 of every run. The two zero entries for H-MLM at $N = 10$ are the runs that fail to learn, and they account for most of the high standard deviation reported for that setting in Table II.

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TABLE III
Configuration. Top: shared. Middle: GlocalDoc. Bottom: H-MLM and fine-tuning.

Setting	Value
Backbone	DistilRoBERTa-base (6 layers, 768)
Max paragraphs / chunk size	50 / 510 tokens
Precision / optimiser	bfloat16 / AdamW
Gradient clip	1.0
GlocalDoc epochs / eff. batch	5 / 8
Encoder LR / weights LR	3×10^{-5} / 1×10^{-2}
Schedule / warm-up	linear / 10%
EMA rate τ	0.996
β final / warm-up fraction	1.0 / 0.25
Free bits λ / latent dim	0.05 nats / 256
Projector / predictor	256-512-768 / 768-512-768
Variance / covariance weight	5.0 / 0.04
Variance hinge γ	0.5
Temporal buffer / var / cov	16 / 25.0 / 1.0
Centre momentum	0.9
Sentence / span / paragraph mask	0.20 / 0.15 / 0.30
H-MLM epochs / eff. batch	3 / 4
H-MLM LR / loss weight	5×10^{-5} / 0.5
H-MLM mask rate	0.15
Fine-tune epochs / LR	10 / 2×10^{-5}
Fine-tune schedule / threshold	cosine, 10% / 0.5
N / seeds	{10, 50, 100} / {0,...,4}

TABLE IV
Per-seed macro-F1 on the test split.

Method	N	s0	s1	s2	s3	s4
No pre.	10	0.000	0.000	0.000	0.001	0.003
	50	0.017	0.033	0.031	0.027	0.016
	100	0.037	0.027	0.037	0.063	0.047
H-MLM	10	0.012	0.000	0.001	0.028	0.054
	50	0.265	0.287	0.291	0.178	0.344
	100	0.390	0.376	0.395	0.317	0.383
GlocalDoc	10	0.042	0.035	0.040	0.049	0.057
	50	0.093	0.094	0.151	0.103	0.101
	100	0.184	0.191	0.178	0.188	0.177

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